Factor Detective: Puzzles with Factors and Multiples Answer Key

Grades 4–5 Operations and Algebraic Thinking

Quick Answers

1. 6	4. —	7. 60	10. 24, 24
2. 9	5. 24, 24	8. 31	11. 13
3. 17	6. 63	9. 36	12. 49

Curriculum

Level: Grades 4–5 Operations and Algebraic Thinking
4.OA.B.4 / 4.OA.C.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 16 tagged checks.

1. Complete the factor pair $4 \times \_\_\_\_ = 24$.

Step 1. A factor pair is two numbers that multiply to the target, so the question is really "four groups of what make 24?"
Step 2. Count by fours: $4, 8, 12, 16, 20, 24$ — six fours, or $24 \div 4$.

$\boxed{6}$

Teaching note: the full pair list for 24 is 1×24, 2×12, 3×8, 4×6 — worth writing out once, because it is what makes 24 composite.

2. Is 342 divisible by 3?

Step 1. Long division would answer it, but the digit-sum rule is faster: a number is divisible by 3 exactly when the sum of its digits is.
Step 2. $3 + 4 + 2 = 9$.

$\boxed{9}$

Step 3. 9 is divisible by 3 (it is 3×3), so 342 is divisible by 3 as well. *Yes.*
Check if you like: $342 \div 3 = 114$ ✓

3. Ari says 51 is prime. Show that it is composite.

Step 1. Prime means *exactly two* factors, 1 and itself. So one factor pair other than 1×51 settles the question.
Step 2. The digits of 51 add to $5 + 1 = 6$, which is divisible by 3, so 3 is a factor. Then $51 \div 3$ gives the partner.

$\boxed{17}$

Step 3. $51 = 3 \times 17$, so 51 has four factors $(1, 3, 17, 51)$ and is composite.
Why Ari was fooled: odd numbers are simply numbers not divisible by 2. Every prime above 2 is odd, but plenty of odd numbers are not prime.

4. Pattern A (adds 4) against pattern B (adds 6): 5th numbers.

Step 1. Each pattern is a skip-count, so its 5th number is five steps of its own size — no need to write out either list in full.
Step 2. Pattern A: $4 \times 5 = 20$. Pattern B: $6 \times 5 = 30$.
Step 3. 20 is less than 30.

$$\boxed{<}$$

Teaching note: the same reasoning says pattern B's nth number is always larger, because it takes bigger steps from a bigger start.

5. First common multiple of 6 and 8.

Step 1. Skip-count both lists and watch for the first number that appears in both — that is what "common multiple" means.
Step 2. Sixes: $6, 12, 18, 24$. Eights: $8, 16, 24$.
Step 3. 24 appears in both, and nothing smaller does. It is the fourth multiple of 6 and the third multiple of 8.

$$\boxed{6 \times 4 = 24}$$
$$\boxed{8 \times 3 = 24}$$

Listen for: 48 — a common multiple, but not the first one.

6. Is 63 a multiple of 5?

Step 1. Multiples of 5 end in 0 or 5, and 63 ends in 3, so the answer is almost certainly no — the skip-count shows why.
Step 2. Counting by fives passes 60 (5×12) and then jumps to 65. So $5 \times 12 + 3 = 63$.

$$\boxed{63}$$

Step 3. 63 sits 3 past a multiple of 5 and 2 short of the next one, so it is *not* a multiple of 5. *No.*
Teaching note: the leftover 3 is the remainder — the same 3 appears in $63 \div 5 = 12$ remainder 3.

7. Write 60 as a product of primes.

Step 1. Split 60 into any factor pair, then keep splitting any factor that is still composite until only primes are left.
Step 2. $60 = 6 \times 10$; then $6 = 2 \times 3$ and $10 = 2 \times 5$.
Step 3. Collect the leaves in order: $2, 2, 3, 5$.

$$\boxed{2 \times 2 \times 3 \times 5 = 60}$$

Worth saying: starting with $60 = 4 \times 15$ or $60 = 5 \times 12$ gives exactly the same four primes. That is not luck — every number has one prime recipe.

8. Pattern $3, 7, 11, 15, \ldots$ — the 8th number.

Step 1. Find the step size first: $7 - 3 = 4$, and the same jump takes 11 to 15, so the pattern adds 4 each time.
Step 2. Getting from the 1st number to the 8th takes 7 jumps, not 8 — the first number is already there.
Step 3. $3 + 7 \times 4 = 3 + 28 = 31$.

$$\boxed{31}$$

Listen for: 35, from taking eight jumps instead of seven.

9. Riddle: in the thirties, divisible by 3 and by 4.

Step 1. A number divisible by both 3 and 4 must be a multiple of $3 \times 4 = 12$, so only the multiples of 12 need checking.
Step 2. Multiples of 12: $12, 24, 36, 48$. The only one in the thirties is 36.
Step 3. Check it: $36 = 3 \times 12$ and $36 = 4 \times 9$, so both divisions come out whole.

$$\boxed{36}$$

Why nothing else works: 30 and 33 and 39 are divisible by 3 but not by 4; 32 is divisible by 4 but not by 3.

10. Bells every 8 minutes and every 12 minutes.

Step 1. Each bell rings at its own multiples, so they meet again at the first common multiple of 8 and 12.
Step 2. Eights: $8, 16, 24$. Twelves: $12, 24$. They meet at 24.

$$\boxed{\text{24 minutes}}$$

Step 3. Check both: $8 \times 3 = 24$ and $12 \times 2 = 24$, so both bells really do ring at minute 24.
Listen for: 96, from multiplying 8×12. That is *a* common multiple, but the bells meet sooner.

11. Ravi says 91 is prime.

Step 1. To disprove a prime claim, one factor pair is enough, so test the small primes in turn: 2 (no, odd), 3 (digits add to 10, no), 5 (no, ends in 1), then 7.
Step 2. $91 \div 7 = 13$, so 7 is a factor and its partner is 13.

$$\boxed{13}$$

Step 3. $91 = 7 \times 13$, so 91 is composite.
Model answer for (b): being odd only means 2 is not a factor — it says nothing about $3, 5, 7, 11, \ldots$ You have to keep testing primes until the test prime multiplied by itself passes the number: $7 \times 7 = 49$ and $11 \times 11 = 121$, so for 91 the testing stops after 7 works (and would have stopped at 11 if it had not).
Manual review: part (b) is the written half.

12. Challenge: exactly three factors, the middle one is 7.

Step 1. The three factors are 1, 7, and the number itself. Factors come in pairs that multiply to the number, and 7 must pair with something.
Step 2. If 7 paired with a different number, that number would be a fourth factor. So 7 has to pair with itself: the number is 7×7.

$$\boxed{49}$$

Step 3. Check: the factors of 49 are $1, 7, 49$ — exactly three.
Model answer for (b): every factor pairs with a partner, so factors usually come two at a time and the count is even. The count is odd only when one factor is its own partner, i.e. the number is a square. And if that square root had any factor other than 1 and itself, the number would pick up extra factors — so the root must be prime. Exactly three factors means prime $\times$ the same prime.
Manual review: part (b) is the written half; $4 = 2 \times 2$ and $25 = 5 \times 5$ are good examples to check the reasoning against.